

A Test Paper on Connectivity

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Table 1: Participant characteristics.

Group	N	Age
Active	21	42.1
Sham	19	44.3

Abstract

We tested 40 participants and found that frontal connectivity decreased ($t = -2.10$, $p = 0.041$).

1 Introduction

Stimulation changes connectivity [2]. Respiration matters too [1]. As shown in Figure 1 and Table 1, the effect is robust. The model is $y_{ij} = \beta_0 + \beta_1 t_{ij} + u_i + \epsilon_{ij}$ with random intercepts.

2 Methods

2.1 Participants

Forty-two participants were recruited from the clinic. Each completed six scanning sessions over five days of treatment.

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \quad (1)$$

3 Results

Connectivity decreased by 0.010 per day in the active arm (Section 1).

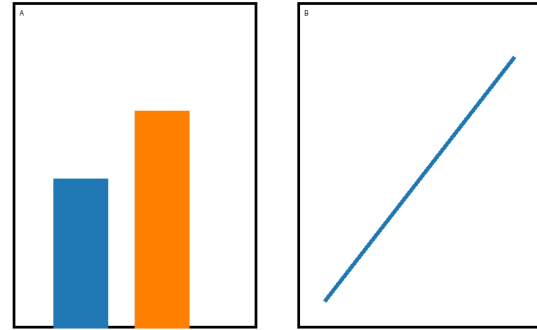


Figure 1: Connectivity by day. A) Active arm. B) Sham arm.

4 Discussion

Effects were modest.

References

- [1] Kim Lee. Respiration and fmri. *NeuroImage*, 230:117–130, 2021.
- [2] Jane Smith and John Doe. Connectivity changes after stimulation. *J Neurosci*, 40:1–10, 2020.

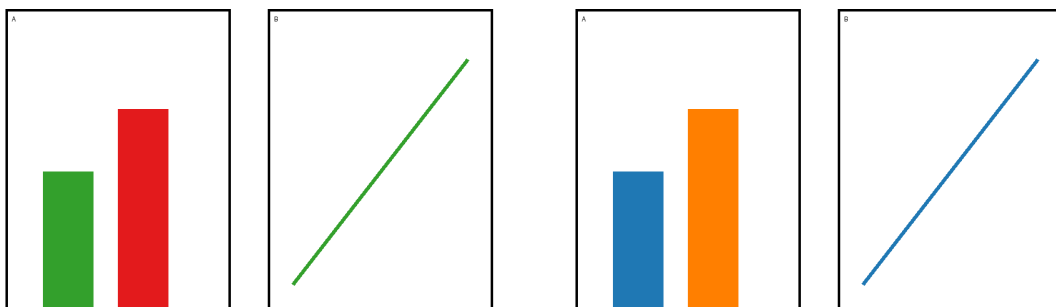


Figure 2: Two subfigures side by side.